

UBT515: BIOSTATISTICS

L	T	P	Cr
2	0	0	2.0

Course Objective: Students will learn about descriptive and inferential statistics and its application in biological data analysis.

Syllabus

Descriptive Statistics: Biology and statistics, Variables and data, Sampling and sampling errors in biological data, Sampling techniques, measures of central tendency, measures of dispersion and variability, Permutations and combinations, Probability, addition and multiplication of probabilities, normal, binomial and Poisson distributions, Binomial and Poisson probabilities, prior probabilities, posterior probabilities and Bayes' theorem.

Hypothesis Testing: Test of hypotheses, one and two sample analysis, Paired sample analysis, Nonparametric statistics and limitations. Confidence limits and tests of confidence, Single, Two and Multifactorial analysis, Non-parametric Analysis of Variance (Kruskal-Wallis test), Multiple comparison tests – Tukey, Newman Keul, Dunnett's test, Scheffe's tests, Contingency tables, Chi-square goodness of fit test.

Regression and Correlation Analysis: Concept of least squares, Simple linear regression, residual sum of squares, regression coefficients, covariance, Pearson coefficient of correlation, coefficient of determination, hypothesis about correlation coefficient, Rank correlation.

Design of Experiments and Data Presentation: Blocking factors, Latin square design, Factorial experiments, Response Surface Methods, Survivorship curves, Graph plotting and significance of Curves, Data representation.

Course Learning Objectives (CLO)

The students will be able to:

1. Classify various types of data and apply basic statistical concepts such as measure of central tendencies, measure of dispersion and sampling.
2. Apply concepts of probability, probability laws, probability distributions and apply them in solving biological problems and statistical analysis.
3. Perform statistical hypothesis testing using tools such as t-test, ANOVA, Chi-square test.
4. Design experiments and solve problems based on relationships among multiple variables.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Text Books

1. Zar, J. H., Biostatistical Analysis. Pearson Education (2014) 5th Edition.
2. Banerjee, B., Mahajan's Methods in biostatistics for medical students and research workers. Jaypee Brothers Medical Publishers (2018) 9th Edition.
3. Montgomery, D. C., Design and Analysis of Experiments. John Wiley & Sons, Inc. (2009) 7th Edition.

Reference Books

1. Rao, K. V., Biostatistics – A Manual of Statistical Methods for Use in Health, Nutrition and Anthropology. Jaypee Brothers (2009) 2nd Edition.
2. Daniel, W. W. & Cross, C. L., Biostatistics: A Foundation for Analysis in the Health Sciences, Wiley Series in Probability and Statistics (2013).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include Assignments/Quizzes)	30

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